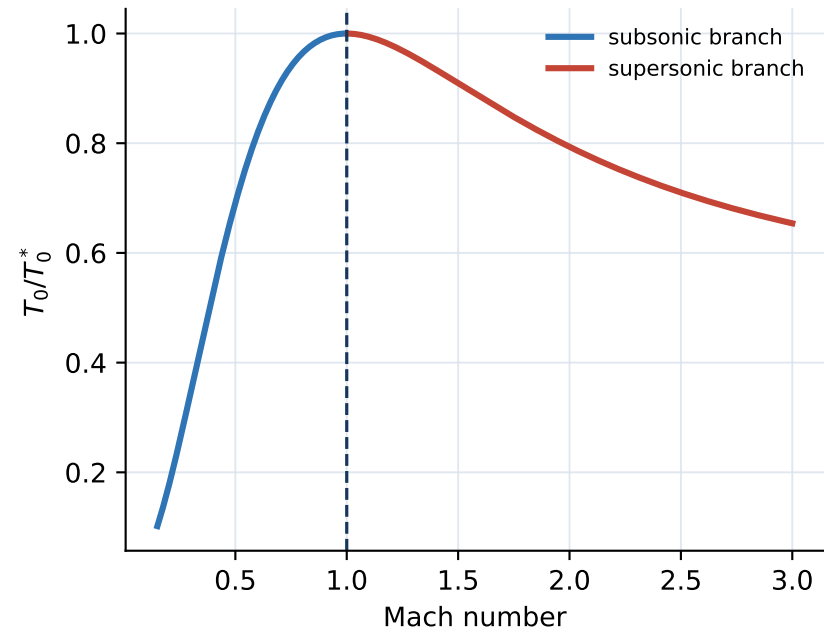
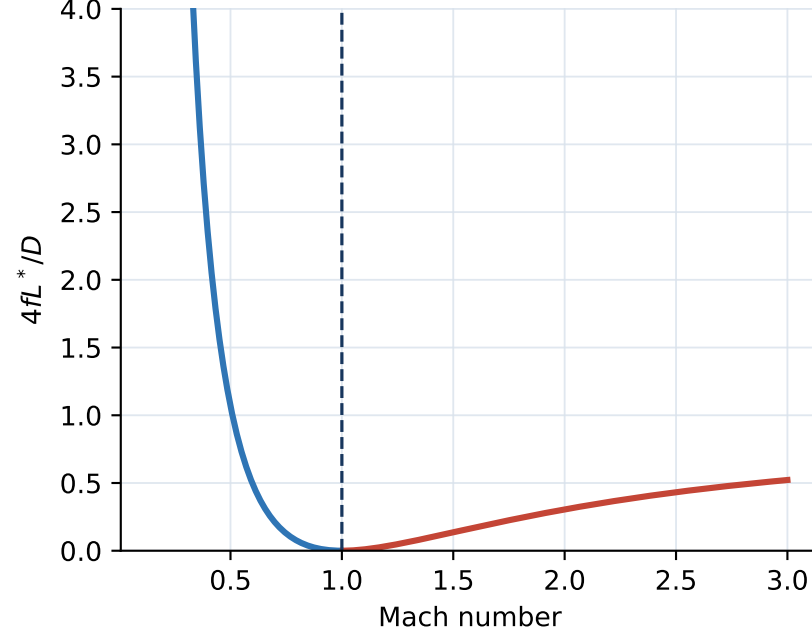


# Week 8: exact gas-dynamics maps before scientific machine learning

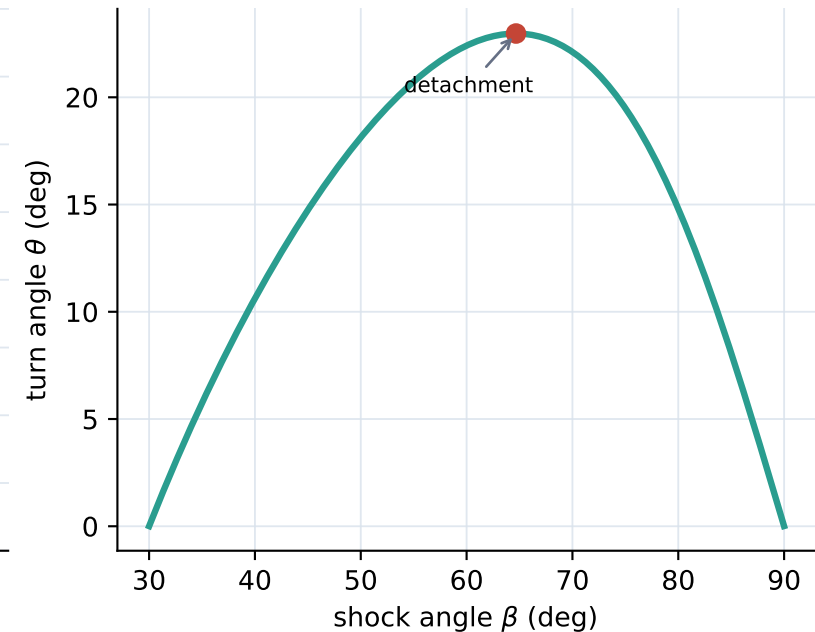
## A Rayleigh heat-addition branches



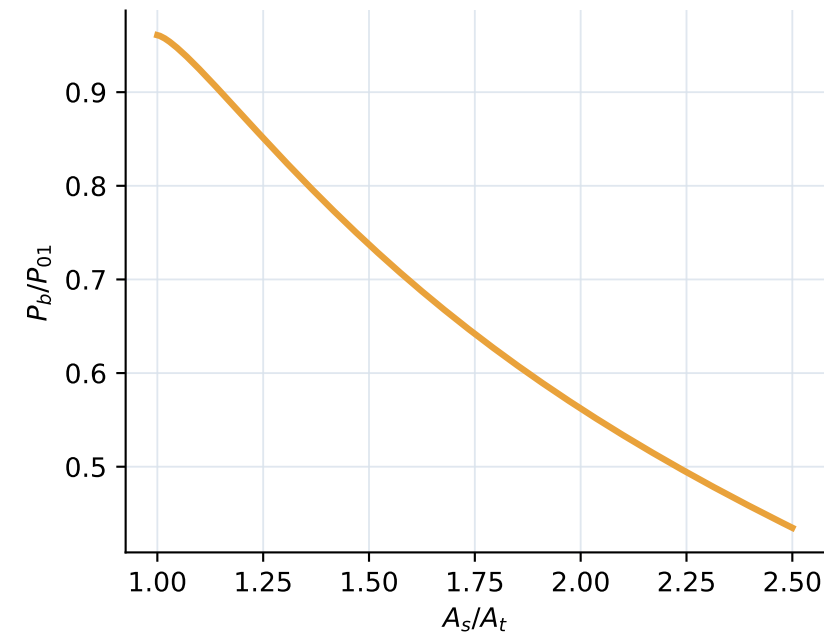
## B Fanno friction-to-choking



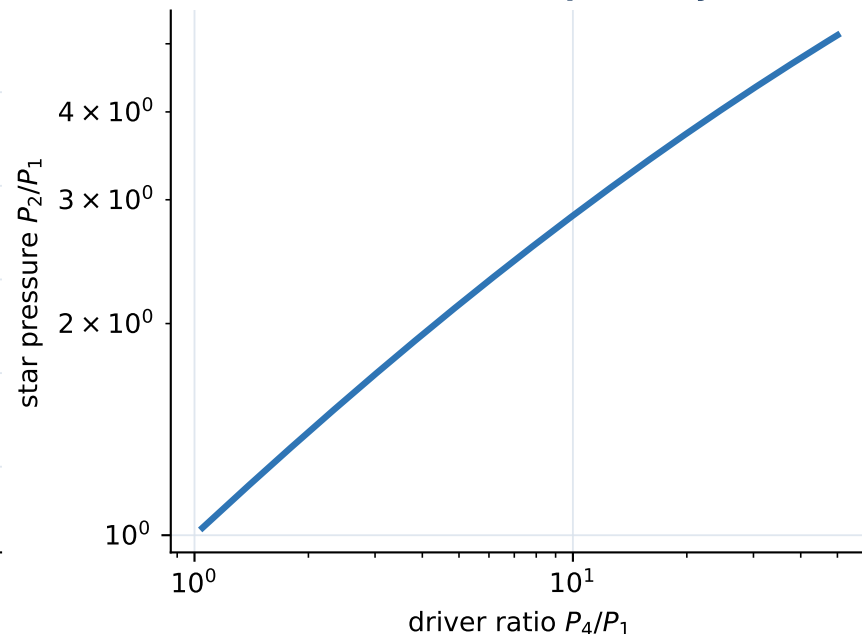
## C Oblique shock, $M_1 = 2$



## D Normal shock in a C-D nozzle



## E Ideal shock-tube compatibility



## F Area-Mach ambiguity

